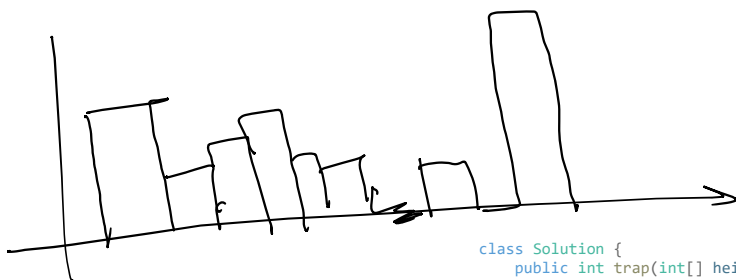


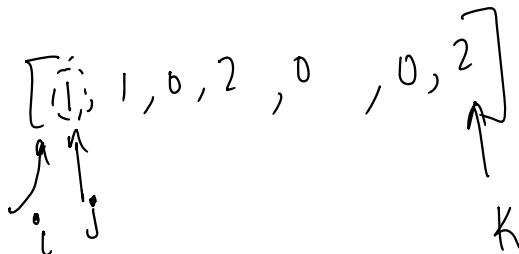
Arrays 18

07 May 2026 13:21



```
class Solution {
    public int trap(int[] height) {
        int n=height.length;
        int [] left_max=new int[n];
        int [] right_max=new int[n];
        left_max[0]=height[0];
        for(int i=1;i<n;i++){
            left_max[i]=Math.max(left_max[i-1],height[i]);
        }
        right_max[n-1]=height[n-1];
        for(int i=n-2;i>=0;i--){
            right_max[i]=Math.max(right_max[i+1],height[i]);
        }
        int total=0;
        for(int i=0;i<n;i++){
            int h=Math.min(left_max[i],right_max[i])-
            height[i];
            total+=h;
        }
        return total;
    }
}
```

* Algorithm → Dutch national flag.



Case 1 :-
 $arr[j] == 0$:
 swap($arr[i], arr[j]$)
 $i++$
 $j++$

Case 2 :-
 $arr[j] == 1$:
 $j++$

Case 3 :-
 $arr[j] == 2$:
 swap($arr[j], arr[k]$)
 $k--$

```
class Solution {
    public void sortColors(int[] nums) {
        int n=nums.length;
```

```

int i=0,j=0,k=n-1;
while(j<=k){
    if(nums[j]==0){
        int temp=nums[i];
        nums[i]=nums[j];
        nums[j]=temp;
        j++;
        i++;
    }
    else if(nums[j]==1){
        j++;
    }
    else if(nums[j]==2){
        int temp=nums[k];
        nums[k]=nums[j];
        nums[j]=temp;
        k--;
    }
}
}
}
}

```

Binary Search:

[1, 2, 3, 4, 5] target = 2

// Template for binary search!!

```

int s=0, e=n-1;
while (s<=e){
    int mid=(s+e)/2;
    if (arr[mid]==target) return mid;
    else if (arr[mid]>target) e=mid-1;
    else s=mid+1;
}
return -1;

```